



EXCEL • POWER PIVOT • DAX • DATA ANALYSIS

Hospital Emergency Room Analysis Dashboard

Problem Statement & Project Requirements Document for an end-to-end Emergency Room reporting dashboard, built to help hospital stakeholders monitor patient flow, wait times, and satisfaction in real time.

Power Query

Power Pivot

DAX

PivotTables

Dashboard Design

1 Purpose of the Project

We need to create a **Hospital Emergency Room Analysis Dashboard** to improve efficiency and provide useful insights. This dashboard will help stakeholders monitor, analyze, and make better decisions for managing patients and improving services — turning raw admission records into a single, self-service reporting view for hospital management.

2 Project Workflow / Steps

- 1 **Prepare Raw Data** — Import and clean the raw Emergency Room dataset using Power Query.
- 2 **Data Modeling** — Build the data model in Power Pivot, relating the fact table to a Calendar dimension table.
- 3 **DAX Calculations** — Create calculated columns (Age Group, Patient Attend Status) using DAX formulas.
- 4 **Pivot Table Analysis** — Summarize patient data using PivotTables across admissions, gender, age, and department referrals.
- 5 **Dashboard Creation** — Assemble KPI cards, charts, and slicers into one interactive dashboard sheet.

3 KPI's Requirement



TOTAL NO. OF PATIENTS



AVERAGE WAIT TIME



PATIENT SATISFACTION
SCORE

4 Charts to Create

► **Patient Admission Status** — Show how many patients were admitted vs. not admitted.

► **Patient Age Distribution** — Group patients by age.

► **Timeliness** — Measure the percentage of patients seen within 30 minutes.

► **Gender Analysis** — Display the number of patients by gender.

► **Department Referrals** — Check which departments patients are referred to the most.

5 Calendar Table Formula

A calendar dimension table is generated in Power Query to enable proper time-intelligence (year / month filtering) across the dashboard.

POWER QUERY M FORMULA

```
= List.Dates(#date(2023,01,01), 731, #duration(1,0,0,0))
```

6 DAX Formulas

DAX FORMULA FOR AGE GROUP

```
= IF([Patient Age]>=70,"70-79", IF([Patient Age]>=60,"60-69", IF([Patient Age]>=45,"45-59", IF([Patient Age]>=30,"30-44", IF([Patient Age]>=15,"15-29", IF([Patient Age]>=5,"05-14","0-4"))))))
```

DAX FORMULA FOR PATIENT ATTEND STATUS

```
= IF([Patient Waittime]<30, "Within Time", "Delay")
```

7 Final Dashboard (Expected Output)

The final deliverable is a single, interactive dashboard sheet in Excel featuring KPI cards, year/month slicers, and visual breakdowns of patient admission status, age distribution, gender split, timeliness, and department referrals.

